Huawei BAS-ACC-37 Routing Protocol Flap Resolution SOP

Node: BAS-ACC-37 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_5959

Description

The occurrence of a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on Huawei's Access Node BAS-ACC-37 at the Basavanagudi POP is primarily attributed to the inherent complexity of the routing protocols involved and the potential for misconfiguration or hardware malfunction. In the context of this device, BGP route churn can be caused by issues such as routing policy misconfiguration, BGP neighbor adjacency loss, or BGP session establishment failures. Similarly, OSPF adjacency loss can be a result of OSPF neighbor state transitions, OSPF authentication issues, or OSPF interface state changes. ISIS LSP flooding can be triggered by ISIS link-state advertisements (LSAs) becoming stale or corrupted, leading to a cascade of LSP flooding across the network. The Severity of this issue is classified as Major due to its potential to cause widespread network instability and service disruptions.

Pre-requisite Checks

- Check: Verify the status of redundant links and ensure they are operational.

- Check: Dispatch remote hands to the Basavanagudi POP to inspect the hardware and perform a visual check for any signs of physical damage or malfunctions.

Potential Root Causes

Cause	Probability
BGP route churn due to routing policy misconfiguration.	High
OSPf adjacency loss due to OSPF neighbor state transitions.	Medium
ISIS LSP flooding due to stale or corrupted ISIS LSAs.	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp all, Expected Output: * BGP table version 12 * BGP local router ID: 192.168.1.1 * BGP table version 12 * BGP local router ID: 192.168.1.1, Description: This command will display the BGP routing table, including all BGP peering sessions, BGP neighbors, and BGP route churn information.

Command: display ospf neighbor, Expected Output: * OSPF neighbor table version 1 * OSPF neighbor table version 1, Description: This command will display the OSPF neighbor table, including OSPF neighbor state transitions and OSPF adjacency loss information.

Command: display isis database, Expected Output: * ISIS database version 12 * ISIS database version 12, Description: This command will display the ISIS link-state database, including ISIS LSAs and LSP
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flooding information.
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Resolution Steps

Step 1: Step: Clear BGP routing table., Command: clear bgp all, Description: This command will clear the BGP routing table, resolving any BGP route churn issues.

Step 2: Step: Re-establish OSPF neighbor adjacency., Command: ospf restart-neighbor, Description: This command will re-establish OSPF neighbor adjacency, resolving any OSPF adjacency loss issues.

Step 3: Step: Clear ISIS LSP flooding., Command: clear isis lsp flooding, Description: This command will clear ISIS LSP flooding, resolving any ISIS LSP flooding issues.

Post-Check Verification

```
Command: display bgp all, Expected Output: * BGP table version 12 * BGP local router ID: 192.168.1.1 * BGP table version 12 * BGP local router ID: 192.168.1.1, Description: This command will verify that BGP route churn has been resolved.

Command: display ospf neighbor, Expected Output: * OSPF neighbor table version 1 * OSPF neighbor table version 1, Description: This command will verify that OSPF adjacency has been re-established.

Command: display isis database, Expected Output: * ISIS database version 12 * ISIS database version 12, Description: This command will verify that ISIS LSP flooding has been cleared.
```

Roll-back Procedures

- Step: Restore original BGP routing table., Command: copy run start, Description: This command will restore the original BGP routing table, resolving any issues caused by the previous resolution steps.

- Step: Re-establish OSPF neighbor adjacency., Command: ospf restart-neighbor, Description: This command will re-establish OSPF neighbor adjacency, resolving any issues caused by the previous resolution steps.

- Step: Clear ISIS LSP flooding., Command: clear isis lsp flooding, Description: This command will clear ISIS LSP flooding, resolving any issues caused by the previous resolution steps.